■ TerraSpectra (Project 1): Professional Engineering & Research Master Blueprint

Enterprise Architecture, Spectroscopy Science, Data Acquisition & Full Module Guide

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■ EXECUTIVE SUMMARY & SYSTEM OVERVIEW

TerraSpectra is an enterprise-grade, GPU-accelerated Precision Agriculture and Geospatial Computer Vision platform.

The Engineering Challenge

Conventional crop health monitoring systems utilize 3-channel RGB (Red, Green, Blue) or 4-channel Multispectral imagery (e.g., NDVI using Near-Infrared). These systems operate on a major flaw: **they can only detect crop diseases after leaves visually turn yellow or brown (chlorosis and necrosis)**. By the time visual symptoms appear to human eyes or standard 2D Convolutional Neural Networks (2D-CNNs), internal cellular structures have already collapsed, making harvest yield loss unrecoverable.

Furthermore, standard 2D-CNN architectures treat image input as flat 2D spatial matrices with a few color channels. They are fundamentally incapable of modeling the **3D spatial-spectral volume** of Hyperspectral Data Cubes, which contain hundreds of narrow, contiguous spectral bands.

The TerraSpectra Solution

TerraSpectra ingests **3D Hyperspectral Data Cubes** capturing **200+ contiguous spectral bands of light** across $400\text{nm} - 2500\text{nm}$. By coupling **3D Convolutional Neural Networks (3D-CNNs)** with **Spectral-Attention Vision Transformers (ViT)**, TerraSpectra detects sub-clinical biochemical anomalies—such as subtle chlorophyll-a/b depletion, anthocyanin spikes, and foliar water stress—predicting fungal blight outbreaks **up to 3 weeks before any visible symptoms manifest on the leaves**.

■ SECTION 1: DOMAIN & PHYSICS RESEARCH (WHAT TO STUDY)

1.1 Electromagnetic Spectrum Physics for Plant Canopy Analysis

Hyperspectral Imaging (HSI) measures light reflectance across hundreds of narrow, contiguous spectral channels (typically $5\text{nm} - 10\text{nm}$ bandwidths).

- **Visible Spectrum (VIS: $400\text{nm} - 700\text{nm}$):**

- **Blue ($400\text{nm} - 500\text{nm}$):** Absorbed strongly by Carotenoids and Chlorophyll-a/b.
- **Green ($500\text{nm} - 600\text{nm}$):** Reflection peak ($\sim 550\text{nm}$), giving healthy plants their green color.
- **Red ($600\text{nm} - 700\text{nm}$):** Chlorophyll-a absorption peak at 675nm.
- **The Red Edge Region ($680\text{nm} - 750\text{nm}$):**
- The boundary between red absorption and Near-Infrared reflection.
- **Critical Disease Signature:** When fungal pathogens (e.g., *Phytophthora infestans*) infect a plant, cell wall damage causes a **leftward shift (blue shift)** of the Red Edge toward shorter wavelengths **weeks before foliar chlorosis is visible**.
- **Near-Infrared (NIR: $750\text{nm} - 1300\text{nm}$):**
- Highly reflected by healthy leaf spongy mesophyll cells ($45\% - 50\%$ reflectance).
- Fungal blight causes internal cell collapse, dramatically dropping NIR reflectance.
- **Short-Wave Infrared (SWIR: $1300\text{nm} - 2500\text{nm}$):**
- SWIR-1 ($1300\text{nm} - 1400\text{nm}$) & SWIR-2 ($1400\text{nm} - 2500\text{nm}$) absorb light based on foliar water content.
- Major atmospheric water vapor absorption bands occur at $1350\text{nm} - 1420\text{nm}$ and $1800\text{nm} - 1930\text{nm}$.

1.2 Radiometric Calibration & Atmospheric Noise Suppression

Satellites measure **Top-of-Atmosphere (TOA) Radiance** (L_{TOA}). Atmosphere contains water vapor (H_2O), carbon dioxide (CO_2), and aerosols that scatter and absorb photons. * **Conversion to Surface Reflectance (ρ_{surface}):** Engineers must study radiative transfer models (such as **6S** - *Second Simulation of a Satellite Signal in the Solar Spectrum* or **FLAASH**): $$\rho_{\text{surface}} = \frac{\pi \cdot (L_{\text{sensor}} - L_{\text{path}})}{\tau_v \cdot (E_0 \cdot \cos(\theta_s) \cdot \tau_s + E_{\text{down}})}$$

■ SECTION 2: DATASETS, FILE FORMATS & ACQUISITION GUIDE

2.1 Standard Data File Formats

1. **HDF5 (`.h5` / `.hdf5`):** Hierarchical Data Format. Stores 3D arrays as datasets alongside attributes (CRS, Bounding Box, Sensor Type).
2. **GeoTIFF (`.tif` / `.tiff`):** Raster format with GeoKeys specifying CRS and Affine Transform matrix mapping pixel (x, y) to real GPS coordinates.
3. **ENVI Header Format (`.hdr` + `.dat`/`.img`):** ASCII header file specifying samples, lines, bands, interleave mode (`BSQ`, `BIL`, `BIP`), and wavelength values.

2.2 Open-Source Hyperspectral Satellite Data Sources

Sensor / Mission	Agency / Source	Bands	Resolution	Data Access Link

AVIRIS / AVIRIS-NG	NASA JPL	224 bands ($380-2510\text{nm}$)	$4\text{m} - 20\text{m}$ airborne	aviris.jpl.nasa.gov
EO-1 Hyperion	NASA / USGS	242 bands ($357-2576\text{nm}$)	30m spaceborne	earthexplorer.usgs.gov
PRISMA	ASI (Italian Space Agency)	240 bands ($400-2500\text{nm}$)	30m spaceborne	prisma.asi.it
EnMAP	DLR (German Aerospace)	224 bands ($420-2450\text{nm}$)	30m spaceborne	enmap.org
Indian HySI	ISRO	64 bands ($400-950\text{nm}$)	500m spaceborne	bhuvan.nrsc.gov.in

■ SECTION 3: MATHEMATICAL & ALGORITHMIC FORMULAS

3.1 Principal Component Analysis (PCA) Math

1. **Flatten 3D Tensor:** $C \in \mathbb{R}^{H \times W \times B} \longrightarrow X \in \mathbb{R}^{N \times B}$ where $N = H \cdot W$.
2. **Mean Centering:** $\bar{X} = X - \mu$.
3. **Covariance Matrix:** $\Sigma = \frac{1}{N - 1} \bar{X}^T \bar{X} \in \mathbb{R}^{B \times B}$.
4. **Eigendecomposition:** $\Sigma V = V \Lambda \quad (\text{Eigenvalues } \lambda_1 \ge \lambda_2 \ge \dots \ge \lambda_B)$.
5. **Cumulative Explained Variance Ratio:** $\text{EVR}(k) = \frac{\sum_{i=1}^k \lambda_i}{\sum_{j=1}^B \lambda_j} \ge 0.985$.
6. **Projection:** $Z = \bar{X} V_k \in \mathbb{R}^{N \times k} \longrightarrow C_{\text{reduced}} \in \mathbb{R}^{H \times W \times k}$.

3.2 Narrow-Band Vegetation Indices (VIs)

- **Photochemical Reflectance Index (PRI):** $\text{PRI} = \frac{R_{531} - R_{570}}{R_{531} + R_{570}}$
- **Water Band Index (WBI):** $\text{WBI} = \frac{R_{970}}{R_{900}}$
- **Structure Insensitive Pigment Index (SIPI):** $\text{SIPI} = \frac{R_{800} - R_{445}}{R_{800} - R_{680}}$

■ SECTION 4: FULL MODULES & LIBRARIES TO DOWNLOAD (WITH EXPLICIT "WHY")

■ Python Backend & Data Science Libraries

```
pip install rasterio h5py netCDF4 scikit-learn torch torchvision timm captum fastapi uvicorn pydantic scipy
```

Package Name	Installation Command	Technical Justification ("WHY It Is Needed")
`rasterio`	`pip install rasterio`	**Geospatial Raster I/O:** Reads GeoTIFF files, extracts Affine Transform matrices, handles Coordinate Reference Systems (CRS: EPSG:4326/3857), and performs bounding box spatial queries.
`h5py`	`pip install h5py`	**HDF5 File Interface:** Ingests 3D Hyperspectral Data Cubes (`.h5` files from NASA AVIRIS/Hyperion), extracts spectral band metadata attributes, and performs slice indexing without loading full multi-gigabyte datasets into RAM.
`netCDF4`	`pip install netCDF4`	**Network Common Data Form:** Ingests Copernicus Sentinel satellite NetCDF files (`.nc`) containing multi-dimensional environmental and spectral array layers.
`scikit-learn`	`pip install scikit-learn`	**Dimensionality Reduction & Metrics:** Implements `sklearn.decomposition.PCA` and `IncrementalPCA` to compress 200+ spectral bands down to 32 components while retaining >98.5% variance. Also provides classification evaluation metrics (`f1_score`, `roc_auc_score`).
`torch` **(PyTorch)**	`pip install torch`	**Deep Learning Framework:** Implements GPU-accelerated 3D Convolutional layers (`nn.Conv3d`), custom DataLoaders, CUDA tensor math, and autograd backpropagation loops for early blight classification.
`torchvision`	`pip install torchvision`	**Vision Transforms & Datasets:** Handles tensor normalization, data augmentation, spatial transforms, and image tensor preprocessing pipelines.
`timm`	`pip install timm`	**PyTorch Image Models:** Provides pretrained Vision Transformer (ViT) and Swin Transformer backbones adapted for spectral self-attention feature extraction.
`captum`	`pip install captum`	**Explainable AI (XAI):** Implements Integrated Gradients and DeepLIFT to extract feature attribution scores across spectral wavelengths, proving *which exact nanometer bands* triggered the disease alert.

`fastapi`	`pip install fastapi`	**High-Performance Async REST API:** Web framework for the backend prediction server. Provides automatic OpenAPI/Swagger documentation, async endpoint handling, and Pydantic request validation.
`uvicorn`	`pip install uvicorn`	**ASGI Web Server:** Lightning-fast production server running the FastAPI application with multi-worker process management.
`pydantic`	`pip install pydantic`	**Data Validation:** Enforces strict type checking for API JSON payloads, coordinate bounding boxes, and metadata DTOs.
`geopandas`	`pip install geopandas`	**Geospatial Vector Engine:** Handles GeoJSON, Shapefiles, spatial joins, and GIS polygon intersection checks for farm field boundaries.
`scipy`	`pip install scipy`	**Scientific & Spectral Algorithms:** Used for Savitzky-Golay spectral smoothing filters, curve fitting, and optimization algorithms.
`shap`	`pip install shap`	**Shapley Additive Explanations:** Computes game-theoretic feature attributions for model predictions.

■ Node.js & React Frontend GIS Libraries

```
npm install react react-dom @deck.gl/core @deck.gl/layers @deck.gl/geo-layers @deck.gl/react mapbox-gl react
```

Package Name	Installation Command	Technical Justification ("WHY It Is Needed")
`react` & **`react-dom`**	`npm install react react-dom`	**UI Library:** Core React 18 component framework for building the user interface, reactive state management, and component lifecycle control.
`@deck.gl/core`	`npm install @deck.gl/core`	**WebGL GIS Engine:** High-performance WebGL-powered data visualization engine capable of rendering millions of geospatial pixels directly on GPU.
`@deck.gl/layers`	`npm install @deck.gl/layers`	**Core GIS Layers:** Provides `BitmapLayer` and `GeoJsonLayer` to overlay disease probability heatmaps directly on top of satellite basemaps.

`@deck.gl/geo-layers`	`npm install @deck.gl/geo-layers`	**Advanced GIS Tile Layers:** Provides `TileLayer` for rendering pyramid raster map tiles dynamically as the user zooms and pans across farm zones.
`@deck.gl/react`	`npm install @deck.gl/react`	**Deck.gl React Wrapper:** Bridges WebGL Deck.gl canvas rendering with React component state.
`mapbox-gl`	`npm install mapbox-gl`	**Mapbox Web Engine:** Renders 3D satellite topography, terrain elevation maps, and high-resolution global satellite imagery basemaps.
`react-map-gl`	`npm install react-map-gl`	**React Wrapper for Mapbox:** Syncs Mapbox viewport state (`longitude`, `latitude`, `zoom`, `pitch`, `bearing`) seamlessly with React components.
`plotly.js-dist-min` & `react-plotly.js`	`npm install plotly.js-dist-min react-plotly.js`	**Spectral Curve Charting:** Renders interactive 224-band continuous reflectance line charts in the UI sidebar when an analyst clicks on any farm pixel.
`lucide-react`	`npm install lucide-react`	**UI Icons:** Modern SVG icon library for GIS controls, alert badges, layer toggles, and status indicators.
`tailwindcss`	`npm install -D tailwindcss postcss autoprefixer`	**Utility-First CSS:** Rapid styling for dark-mode glassmorphism cards, responsive sidebars, and control panels.
`vite`	`npm install -D vite @vitejs/plugin-react`	**Build Tool & Dev Server:** Next-generation frontend build tool providing instant Hot Module Replacement (HMR) and optimized TypeScript bundling.

■ SECTION 5: 6-PHASE ENGINEERING EXECUTION ROADMAP

■ Phase 1: Research & Data Acquisition

1. Acquire AVIRIS/Hyperion sample datasets or generate synthetic 224-band HDF5 files.
2. Verify spectral reflectance curves ($400\text{nm} - 2500\text{nm}$) for healthy crops vs infected blight zones.
3. Obtain Mapbox API Access Token from mapbox.com.

■ Phase 2: Data Ingestion & Dimensionality Reduction

1. Implement `HyperspectralDataLoader` to parse `.h5` and `.tif` files with `h5py` and `rasterio`.

2. Implement `SpectralPCAReducer` using `sklearn.decomposition.PCA` to reduce 224 bands $\rightarrow 32$ components (>98.5% variance retained).
3. Compute narrow-band Vegetation Indices (PRI, WBI, SIPI).

■ Phase 3: AI Model Engineering (PyTorch 3D-CNN + ViT)

1. Build PyTorch 3D-CNN spatial-spectral feature extractor accepting 5D tensors `(Batch, 1, Components, Height, Width)`.
2. Integrate Vision Transformer (ViT) self-attention module to learn inter-band dependencies.
3. Train model on healthy vs infected spatial-spectral samples.

■ Phase 4: Explainability (XAI) & Validation Audit

1. Implement `Captum` Integrated Gradients to extract wavelength importance scores.
2. Confirm the model's top attribution weights align with the Red Edge shift ($680\text{nm} - 750\text{nm}$).

■ Phase 5: Production FastAPI Tiling API

1. Build sliding-window tiling algorithm ($256 \times 256 \times 32$ tensor chunks with 16px overlapping borders).
2. Wrap model in FastAPI async endpoints (`/api/v1/predict/tile`).

■ Phase 6: Interactive React 18 + Deck.gl 3D GIS Dashboard

1. Scaffold React 18 + Vite + TypeScript application.
2. Build `MapContainer.tsx` integrating Deck.gl `BitmapLayer` / `GeoJsonLayer` over Mapbox GL satellite basemap.
3. Add Plotly.js spectral reflectance chart sidebar and historical timeline slider.